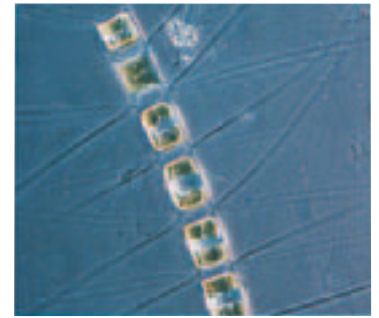


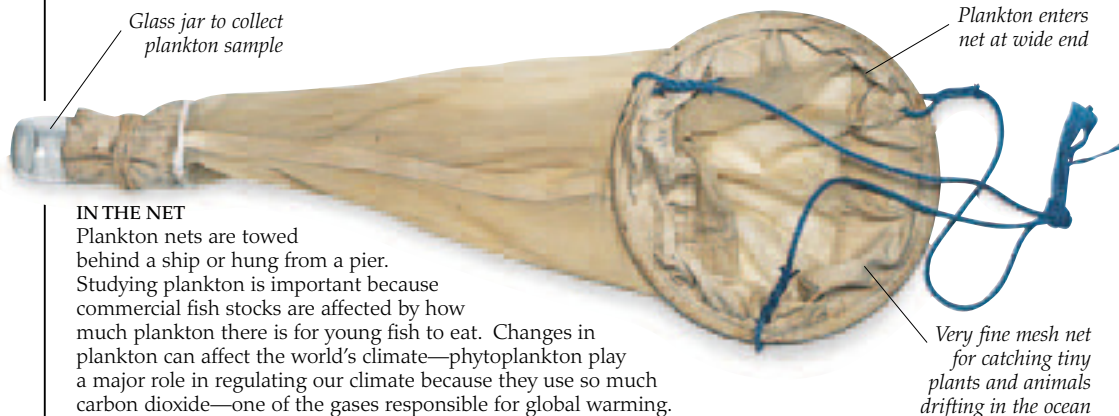
Sea meadows

THE MOST ABUNDANT PLANTS IN THE OCEAN are too small to be seen with the naked eye. Often single-celled, these minute, floating plants are called phytoplankton. Like all plants they need sunlight to grow, so are only found in the ocean's upper zone. With the right conditions, phytoplankton multiply quickly—within a few days—as each cell divides into two, and so on. To grow, phytoplankton need nutrients from seawater and lots of sunlight. The most light occurs in the tropics but nutrients, especially nitrogen and phosphorus, are in short supply, restricting phytoplankton's growth. Spectacular phytoplankton blooms are found in cooler waters where nutrients (dead plant and animal waste) are brought up from the bottom during storms, and in both cool and warm waters where there are upwellings of nutrient-rich water. Phytoplankton are eaten by swarms of tiny, drifting animals (zooplankton), which provide a feast for small fish (such as herring), which in turn are eaten by larger fish (such as salmon), which in their turn are eaten by still larger fish or other predators (such as dolphins). Some larger ocean animals (whale sharks and blue whales) feed directly on zooplankton.



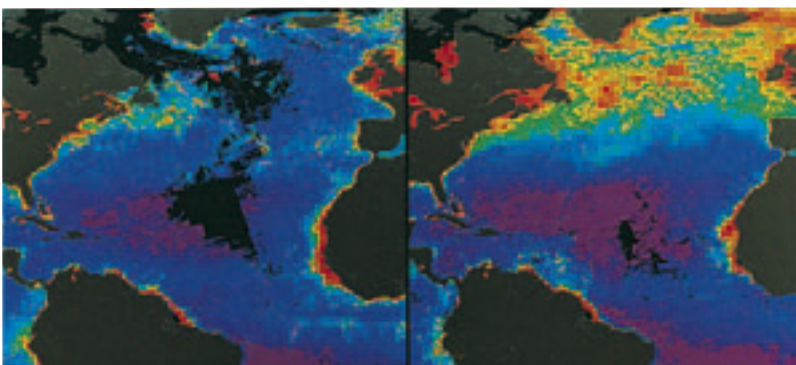
PLANT FOOD

This diatom is one of many phytoplankton that drift in the ocean. Diatoms are the most common kinds of phytoplankton in cooler waters, but dinoflagellates, called single-celled plants, are common in tropical waters. Many diatoms are single cells, but this one consists of a chain of cells.



IN THE NET

Plankton nets are towed behind a ship or hung from a pier. Studying plankton is important because commercial fish stocks are affected by how much plankton there is for young fish to eat. Changes in plankton can affect the world's climate—phytoplankton play a major role in regulating our climate because they use so much carbon dioxide—one of the gases responsible for global warming.



OCEAN IN BLOOM

Images from data collected from a space satellite (*Nimbus 7*) show densities of phytoplankton in the Atlantic Ocean. Red shows where phytoplankton is densest through yellow, green, blue to violet where phytoplankton is least dense. Phytoplankton's spring bloom (right) occurs when days are longer and more nutrients come up from the bottom. A second, smaller bloom of phytoplankton occurs in the fall. When phytoplankton dies, it sinks to the seabed with gelatinous zooplankton remains, making sticky clumps called marine "snow."

Older stage crab larva showing pincers

Younger stage crab larva

